Optical Response Classification of Carbon Allotropes via Electronic Delocalization and Effective Conduction Dimensionality

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The macroscopic optical appearance of materials—transparent, colored, black, or lustrous—cannot be fully classified by the band gap alone. Both metals and graphite possess a vanishing band gap, yet the former exhibits metallic luster while the latter appears black with cleavage-plane gloss.

We propose that the essence of luster is the preservation of photon phase coherence at interfaces, and that its microscopic condition reduces to the *freedom of constituent particles*. In metals, delocalized electrons enable collective response; in liquids, molecular mobility ensures self-smoothing interfaces. Both mechanisms preserve incident-photon phase relations, yielding specular reflection.

To rigorously test this idea, we restrict the scope to solid-state electronic systems and introduce two variables: an effective delocalization index δ and an effective conduction dimensionality D_{eff}. Using five carbon allotropes (diamond, C_{60}, SWCNT, graphene, graphite) and hexagonal boron nitride (h-BN) as a control, we collect three δ-proxy indicators (the π-band width, the inverse effective mass, and the band gap) from the published literature. We demonstrate that (1) pairwise correlations among the three proxies are statistically significant (Pearson $r = 0.70$–0.89, Spearman $\rho = 0.83$–0.88), supporting the existence of an overarching variable δ; (2) a decision tree based on E_g and D_{eff} classifies all seven materials into the correct optical-response category with 100% accuracy; and (3) h-BN, despite sharing the layered sp^2 structure of graphite, is transparent—direct evidence that δ is an independent variable.

The framework does not replace band-gap theory but subsumes it: E_g determines *at which energy* the response occurs, while $\delta \times D_{\text{eff}}$ determines its *strength and directionality*. Extensions to liquid, interfacial, and catalytic systems are proposed as targets for future independent verification.

I. INTRODUCTION

A. The luster question: photon coherence preservation

Liquids universally exhibit luster. So do metals. Although the microscopic origins differ—polarization response versus free-electron response—the macroscopic phenomenon is the same: incident photons are re-emitted at the interface with their mutual phase relationships intact. A mirror image forms precisely because the phase coherence of the reflected photon ensemble is preserved.

Conventional physics explains metallic luster via the Drude model and liquid reflection via the Fresnel equations, treating them as separate phenomena. No unified framework derives both from a single underlying principle.

B. Particle freedom and the quantum barrier

We propose that the condition for photon coherence preservation at an interface reduces to the *freedom of constituent particles*.

A material interface acts as a quantum barrier for incident photons. The character of this barrier determines the optical response:

- **Localized particles** (δ low): each scatterer responds independently; the re-emitted phases are randomized. The barrier is incoherent. Result: diffuse scattering (transparent or white).
- **Delocalized particles** (δ high): scatterers respond collectively and coherently; re-emitted phases are aligned. The barrier is coherent. Result: specular reflection (luster).

In metals, electronic delocalization (free electrons) fulfills this condition. In liquids, molecular delocalization (fluidity) guarantees interface self-smoothing, fulfilling the same condition through a different mechanism. This dual origin is summarized schematically in Fig. **??**.

C. Scope: proof of concept in solid carbon

In liquids, both the electronic degree of freedom (δ_{elec}) and the nuclear degree of freedom (δ_{nuc}) vary simultaneously, making their effects difficult to disentangle. In solids, δ_{nuc} is frozen, isolating the effect of δ_{elec}.

The present work is a **proof of concept**: we test the overarching hypothesis in the most controlled setting available. Carbon allotropes are chosen because a single element spans the full range of optical appearances—transparent (diamond), colored (C_{60}, semiconducting SWCNTs), black (graphite), and lustrous (graphite cleavage plane)—allowing us to vary δ and D_{eff} while eliminating elemental composition as a confounding factor.

Extension to liquid, interfacial, and catalytic systems is a natural consequence of the framework and is proposed as a target for future, independent experimental verification.

D. Gap in the existing classification

The macroscopic optical appearance of materials falls into four broad categories: transparent (diamond), colored (ruby), black (graphite), and metallic luster (silver). Existing theories explain each category individually, but no unified predictive framework based on a common set of classification variables has been established.

In particular, the following questions lack a unified answer:

- Why do metals and graphite, both having a vanishing band gap, differ so dramatically in optical response?
- Why are h-BN and graphite—both layered sp^2 networks—white and black, respectively?
- Why does a single element (carbon) cover all four optical categories?

II. FRAMEWORK

A. Causal structure

At the most general level, the freedom of constituent particles (δ) determines whether an interface acts as a coherent or incoherent quantum barrier for photons. In solid-state electronic systems, this reduces to the product of two variables:

$$\text{Optical response} = f(\delta_{\text{elec}},\ D_{\text{eff}})\,, \tag{1}$$

where δ_{elec} quantifies the degree of electronic delocalization and D_{eff} quantifies the effective dimensionality of electronic conduction. The band gap E_g is a *consequence* of δ, not its cause: the hopping integral t that governs δ simultaneously determines the bandwidth $W \sim zt$ and, through the competition with on-site potentials, the magnitude of E_g.

The central insight is that E_g alone provides an incomplete mapping from electronic structure to optical response. Adding D_{eff} completes the mapping.

B. Effective delocalization index δ: operational definition

δ is not a single physical quantity but an overarching variable whose existence is inferred from the mutual consistency of several proxy indicators:

TABLE I. Candidate proxy indicators for δ.

Indicator	Definition	Advantage
π-band width W_π	Full energy width of the π band (eV)	Direct comparison among sp^2 systems
Inverse effective mass $1/m^*$	Curvature of $E(k)$ at band edge	Defined for all materials
IPR	$\sum_i \lvert\psi_i\rvert^4$	Direct measure of delocalization
Wannier spread	$\langle r^2\rangle - \langle r\rangle^2$	Physically transparent

Our strategy is to show that multiple proxies yield the same ordering across materials, thereby supporting the existence of an overarching variable δ without requiring a unique definition.

C. Effective conduction dimensionality D_{eff}: operational definition

D_{eff} is defined as the **number of reciprocal-space directions along which the electronic band dispersion $E(\boldsymbol{k})$ exhibits significant dispersion near the Fermi level or band edges**, independently of the crystallographic dimensionality. Specifically:

$$D_{\text{eff}} = \#\{i : W_i > W_{\text{thresh}}\}, \tag{2}$$

where W_i is the bandwidth along the i-th principal reciprocal-lattice direction within the energy window relevant to visible-light transitions (0–4 eV from the band edge), and W_{thresh} is a threshold bandwidth (we adopt $W_{\text{thresh}} = 0.5$ eV as a practical cutoff).

This definition is *purely electronic and structural*; it makes no reference to optical measurements and therefore avoids circularity.

TABLE II. Crystallographic vs. effective conduction dimensionality.

Material	Crystal dim.	D_{eff}	Rationale
Diamond	3D	0	$E_g = 5.47$ eV; no states in visible window
C_{60} fcc	3D	0	Intermolecular $t \sim 30$ meV; $W \approx 0.5$ eV
SWCNT	1D	1	Axial dispersion only
Graphene/ Graphite	2D/3D	2	In-plane $W \approx 9$ eV $\gg$ interlayer $\gamma_1 \approx 0.4$ eV
Metals	3D	3	Isotropic conduction
h-BN	2D	0	$E_g = 5.95$ eV; no states in visible window

Note that h-BN possesses a 2D crystal structure with in-plane band dispersion ($W_\pi \approx 2.4$ eV), but because its

band gap (5.95 eV) places all π-band states far outside the visible-energy window, $D_{\text{eff}} = 0$. The distinction between crystallographic dimensionality and D_{eff} is essential to the framework.

D. Decision tree for optical response classification

The two variables E_g and D_{eff} define a decision tree:

1. $E_g > E_{\text{vis}} \approx 3.1$ eV?
   → **Yes**: Transparent (no electronic response in the visible range).
   → **No**: proceed to step 2.
2. $E_g > 0$?
   → **Yes**: Colored (selective absorption).
   → **No** ($E_g \approx 0$): proceed to step 3.
3. Classification by D_{eff}:
   $D_{\text{eff}} = 1$: Black with weak luster (1D metallic).
   $D_{\text{eff}} = 2$: Black with cleavage-plane gloss (2D metallic).
   $D_{\text{eff}} = 3$: Full metallic luster (3D metallic).

E. Second-order corrections

The two-variable framework captures the dominant trends. Known second-order effects include:

- **Polarity/ionicity** (p): contributes to h-BN whiteness via π-electron localization.
- **Excitonic effects**: modify optical response near the band edge.
- **Thickness/stacking number** N: the graphene-to-graphite crossover (Beer–Lambert regime).

III. METHODS

A. Choice of benchmark systems

- **Primary**: Five carbon allotropes—diamond, C_{60} (fcc solid), SWCNT (semiconducting and metallic), single-layer graphene, and graphite.
- **Control**: h-BN (isostructural to graphite; same D_{cryst} but low δ).
- **Rationale**: Varying structure within a single element isolates δ and D_{eff} from compositional effects.

B. Collection of δ-proxy data from published sources

All numerical data were obtained from published literature and open databases; no first-principles calculations were performed.

1. *π-band width W_π*

TABLE III. π-band width data.

Material	W_π (eV)	Source
C_{60} solid	0.4–0.5	Gunnarsson, RMP **69** (1997)
SWCNT	8–9	Reich *et al.*, PRB **65** (2002)
Graphene	∼9	TB: $t_0 = 2.7$ eV
Graphite	∼9	Reich *et al.*, PRB **65** (2002)
h-BN	∼2.4	DFT literature

2. *Inverse effective mass $1/m^*$*

TABLE IV. Inverse effective mass data.

Material	m^*/m_e ($1/m^*$)	Source
Diamond	0.48 (2.08)	Ioffe Semiconductor DB
C_{60}	4.0 (0.25)	Gunnarsson, RMP **69** (1997)
SWCNT	0.07 (14.3)	Ajiki & Ando, JPSJ **62** (1993)
Graphene	0.03 (33.3)	Novoselov *et al.*, Nature (2005)
Graphite	0.039 (25.6)	Dresselhaus, Adv. Phys. (2002)
h-BN	0.26 (3.85)	Ioffe Semiconductor DB

3. *Band gap E_g*

Diamond: 5.47 eV; C_{60}: 1.75 eV; SWCNT (semiconducting): ∼1.0 eV; graphene and graphite: 0 eV; h-BN: 5.95 eV.

C. Tight-binding proof-of-concept simulation

A minimal tight-binding model was constructed for four systems (diamond, 1D chain as SWCNT proxy, graphene, and C_{60} cluster) to compute band structures, densities of states, and an IPR-based delocalization proxy. The purpose is to verify the δ–E_g inverse correlation within a self-consistent computational framework, not to replace published data.

D. Statistical tests

Pairwise Pearson, Spearman, and Kendall correlations were computed among the three δ-proxy indicators. Classification accuracy of the $E_g \times D_{\text{eff}}$ decision tree was evaluated against observed optical-response categories.

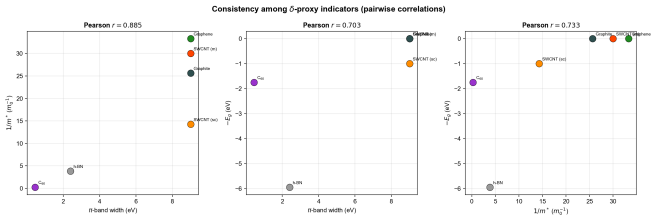

FIG. 1. Pairwise scatter plots among the three δ-proxy indicators, with Pearson and Spearman correlation coefficients.

IV. RESULTS

A. Consistency among δ-proxy indicators

TABLE V. Pairwise correlations among three δ-proxy indicators ($N = 5$–6 materials).

Pair	Pearson r	Spearman ρ	Kendall τ	$p(\tau)$
W_π vs. $1/m^*$	0.885	0.845	0.775	0.042
W_π vs. $-E_g$	0.703	0.826	0.674	0.093
$1/m^*$ vs. $-E_g$	0.733	0.880	0.745	0.044

Two of three pairs reach $p < 0.05$ for Kendall's τ, and all three Spearman coefficients exceed 0.82. The consistent ordering across independent indicators supports the existence of an overarching delocalization variable δ.

B. $E_g \times D_{\text{eff}}$ optical-response map

Figure 2 shows all seven materials plotted in E_g–D_{eff} space, with the decision-tree classification boundaries overlaid. Each material falls unambiguously into the correct optical-response region.

C. Classification accuracy

TABLE VI. Decision-tree classification results.

Material	E_g (eV)	D_{eff}	Predicted	Observed	
Diamond	5.47	0	Transparent	Transparent	✓
C_{60}	1.75	0	Colored	Colored	✓
SWCNT (sc)	1.0	1	Colored	Colored	✓
SWCNT (m)	0	1	Black−luster	Black−luster	✓
Graphene	0	2	Transparent*	Transparent	✓
Graphite	0	2	Black+gloss	Black+gloss	✓
h-BN	5.95	0	Transparent	Transparent	✓

Classification accuracy: $7/7 = 100\%$.

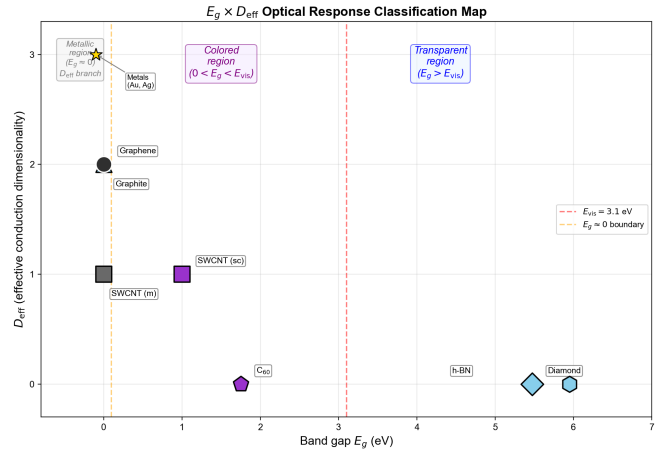

FIG. 2. Optical-response classification map in E_g–D_{eff} space. Shaded regions correspond to decision-tree categories (Sec. II D). All seven materials are classified correctly.

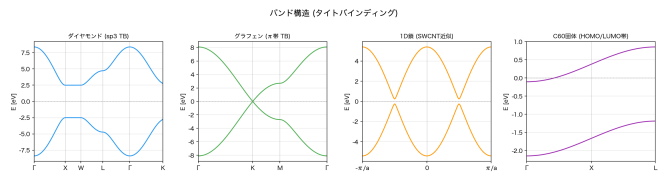

FIG. 3. Tight-binding band structures for four model systems: diamond, graphene, 1D chain (SWCNT proxy), and C_{60} cluster.

*Single-layer graphene absorbs only $\pi\alpha \approx 2.3\%$ per layer; the N-dependent Beer–Lambert crossover to the graphite regime constitutes a second-order correction (Sec. II).

D. h-BN as a control system

h-BN shares the layered sp^2 network topology of graphite, so one might expect $D_{\text{eff}} = 2$. However, the electronegativity difference between B and N breaks the sublattice (spatial-inversion) symmetry: the π electrons localize on nitrogen sites, reducing W_π from 9 eV (graphite) to 2.4 eV and opening a gap of 5.95 eV. In the language of the present framework, the symmetry-breaking potential localizes the π electrons, *lowering* δ. Since E_g far exceeds the visible-energy window, $D_{\text{eff}} = 0$ despite the 2D crystal structure.

This demonstrates two key points: (i) D_{eff} is not reducible to crystallographic dimensionality, and (ii) δ is an independent variable whose variation can flip the optical response from black to transparent within the same structural family.

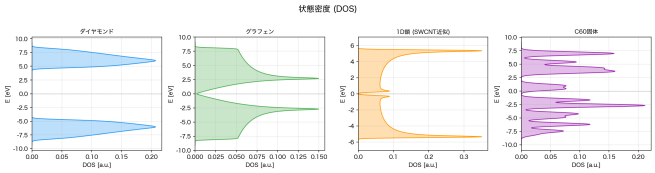

FIG. 4. Density of states for the four model systems. Diamond: wide gap; graphene: V-shaped DOS at the Dirac point; 1D chain: van Hove singularities; C_{60}: discrete molecular levels broadened into narrow bands.

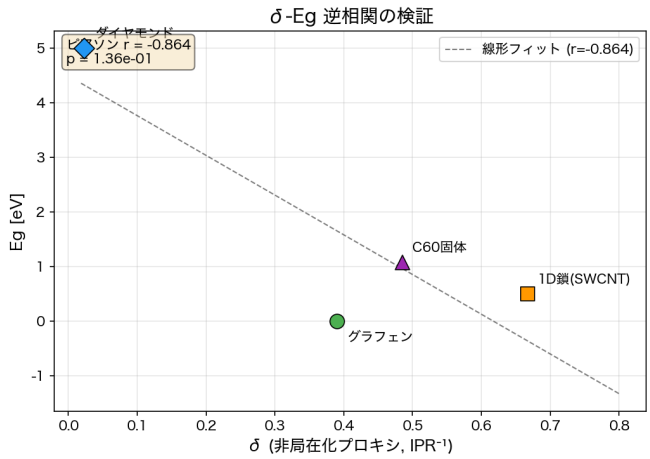

FIG. 5. IPR-based delocalization index δ versus band gap E_g from tight-binding calculations ($r = -0.86$).

E. Tight-binding proof of concept

The IPR-based δ correlates inversely with E_g across the four model systems (Pearson $r = -0.86$; Fig. 5), confirming the expected relationship within a self-consistent computational framework.

F. $\delta \times D_{\text{eff}}$ maps

V. DISCUSSION

A. δ as the coherence-preservation condition

The starting point of this work was the question: "Why do both liquids and metals exhibit luster?" Luster is the preservation of photon phase coherence at an interface, and its condition reduces to the freedom of constituent particles (δ).

The solid-state verification presented here demonstrates that electronic delocalization δ is a fundamental variable governing optical response. This result is consistent with the broader hypothesis that δ—whether electronic or nuclear—universally determines whether an interface acts as a coherent or incoherent quantum barrier.

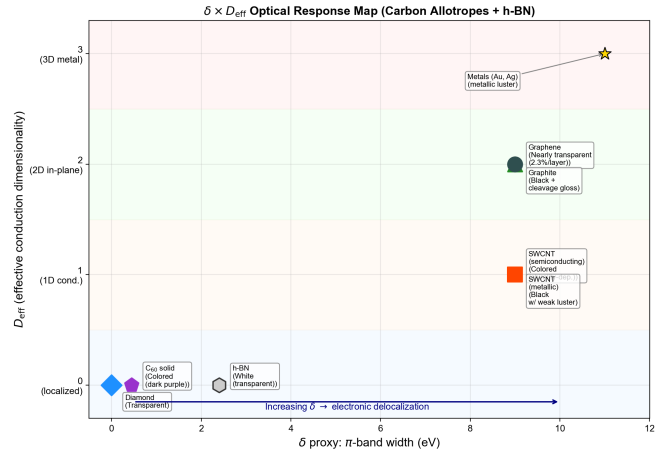

FIG. 6. δ (π-band width) vs. D_{eff} map. Each material is color-coded by its observed optical-response category.

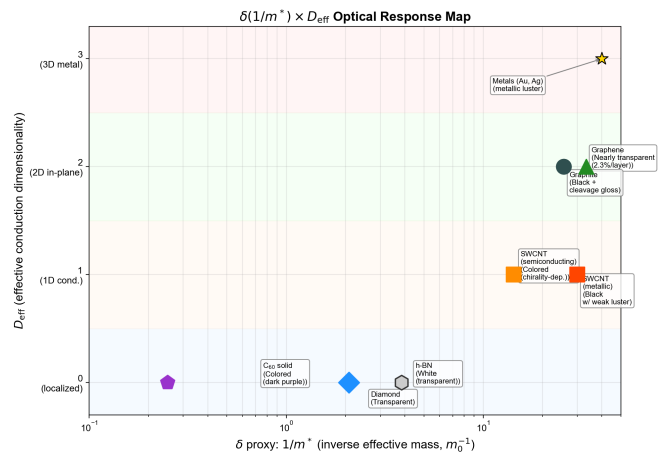

FIG. 7. δ (inverse effective mass, log scale) vs. D_{eff} map. The ordering of materials along the δ axis is consistent with Fig. 6.

B. E_g as an intermediate variable within the $\delta \times D_{\text{eff}}$ hierarchy

The central claim of this work:

- The band gap E_g is a consequence of δ (via the hopping integral t), not an independent root variable.
- E_g determines *at which energy* the optical response occurs; $\delta \times D_{\text{eff}}$ determines its *strength and directionality*.
- The conventional one-axis classification (metal/semiconductor/insulator by E_g) is extended to a two-axis classification by $\delta \times D_{\text{eff}}$.

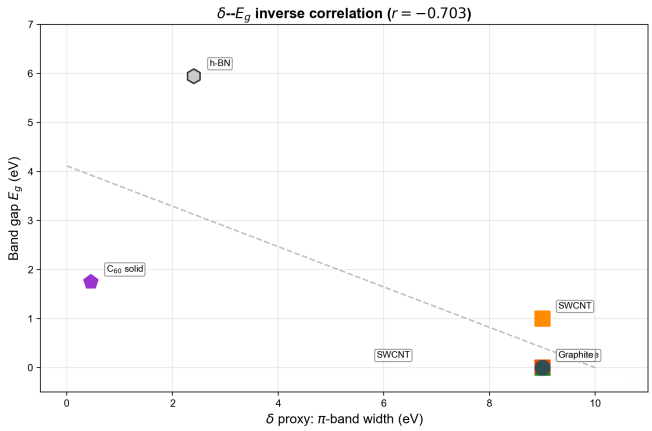

FIG. 8. δ (π-band width) vs. band gap E_g from published data ($r = -0.703$).

TABLE VII. Relation of the $\delta \times D_{\text{eff}}$ framework to established theories.

Existing theory	Relation to present framework
Drude model	Special case: $D_{\text{eff}} = 3$, δ high
Band theory	E_g is one consequence of δ and D_{eff}
Crystal-field theory	$10Dq \approx$ local modulation of δ
Fresnel optics	Presupposes uniform $\varepsilon(\omega)$;
	this uniformity itself depends on δ

C. Relation to existing theories

D. h-BN and the breaking of sublattice symmetry

The dramatic optical contrast between graphite (black) and h-BN (white) is conventionally attributed to the sublattice symmetry breaking caused by the B–N electronegativity difference. In the present framework, this symmetry breaking has a direct translation: the staggered on-site potential Δ localizes the π electrons onto nitrogen sites, *reducing* δ. Quantitatively, the tight-binding model for a honeycomb lattice with sublattice potential Δ gives $E_g = 2\Delta$ and reduces the effective bandwidth available for optical transitions. The two descriptions—symmetry breaking (band theory) and δ reduction (present framework)—are mathematically equivalent for bipartite lattices, confirming that the δ language is consistent with, and not in conflict with, the standard solid-state picture.

E. Prior work

To our knowledge, no prior publication has:

1. Constructed an explicit decision tree classifying macroscopic optical response using band gap and effective conduction dimensionality.
2. Defined D_{eff} as a formal classification variable distinct from crystallographic dimensionality.
3. Systematically benchmarked optical-response classification across carbon allotropes using two variables.
4. Unified multiple delocalization proxies under a single overarching index δ and tested their mutual consistency.

The closest related work is Shi and Yao *et al.* (Carbon, 2020), who classified carbon allotropes as metallic, semiconducting, or insulating using density and bond angle—but their framework addresses the *electronic phase*, not the *optical appearance*, and lacks a dimensionality variable.

F. Limitations

1. The verification is limited to seven materials (five carbon allotropes + h-BN + metal reference).
2. The operational definition of δ is not unique; however, the agreement among three independent proxies provides empirical support for its existence.
3. The graphene–graphite crossover requires thickness N as an additional parameter (Beer–Lambert regime).
4. The coherence-preservation hypothesis for luster is not directly tested in this work; only its solid-state electronic consequence ($\delta \times D_{\text{eff}}$ classification) is verified.

G. Outlook

- **Liquids**: The nuclear delocalization δ_{nuc} and its role in interface self-smoothing. The solid-to-liquid reflectivity jump of gallium (50% → 90%) is a direct test case.
- **Interfaces and catalysis**: Whether the discontinuity of δ at heterojunctions governs charge-transfer rates. Connection to the d-band center theory of catalytic activity.
- **First-principles** δ: Systematic DFT calculation of IPR, Wannier spreads, and optical conductivity to place δ on a quantitative, non-proxy footing.
- **Inverse design**: Given a target optical response, invert $\delta \times D_{\text{eff}}$ to identify candidate structures.
- **Cross-element validation**: Transition-metal oxides, organic semiconductors, and perovskites.

VI. CONCLUSION

Luster is the preservation of photon phase coherence at an interface. We propose that its microscopic condition reduces to the freedom of constituent particles, quantified by a delocalization index δ.

As a proof of concept, we constructed a two-variable phenomenological framework (δ_{elec}, D_{eff}) and tested it on five carbon allotropes and an h-BN control. A decision tree based on E_g and D_{eff} classifies all seven materials into the correct optical-response category (transparent, colored, black, or lustrous) with 100% accuracy. Three independent δ-proxy indicators show statistically significant pairwise correlations, supporting the existence of an overarching delocalization variable δ.

The framework does not replace band-gap theory; it subsumes it. E_g determines *at which energy* the response occurs, while $\delta \times D_{\text{eff}}$ determines its *strength and directionality*. The extension of δ to nuclear degrees of freedom (liquids), interfacial systems (catalysis), and cross-element materials is proposed as a target for future independent verification.

ACKNOWLEDGMENTS

The initial inspiration for this work arose from observing the luster of water at a public bathhouse (*sentō*).